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One-class support vector machines—an application in machine fault detection and classification

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Abstract

Fast incipient machine fault diagnosis is becoming one of the key requirements for economical and optimal process operation management. Artificial neural networks have been used to detect machine faults for a number of years and shown to be highly successful in this application area. This paper presents a novel test technique for machine fault detection and classification in electro-mechanical machinery from vibration measurements using one-class support vector machines (SVMs). In order to evaluate one-class SVMs, this paper examines the performance of the proposed method by comparing it with that of multilayer perception, one of the artificial neural network techniques, based on real benchmarking data.

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Keywords: Machine fault diagnosis; Support vector machines; One-class classification; Artificial neural networks; Multilayer perception

1. Introduction

As the need for effective machine diagnosis system is becoming more urgent with increasing complex, expensive manufacturing facilities, a fast and accurate fault detection and diagnosis technique for manufacturing facilities is considerably important in high technology industry such as semiconductor manufacturing. There have been considerable researches done on fault diagnosis using an artificial neural network, discriminative analysis, expert system, and decision tree. These existing approaches have the different properties as well as advantages and disadvantages. In order to diagnose faults efficiently, we need to learn the previous fault patterns and signal trends. This can improve the accuracy of fault detection and classification, and reduce the cost of maintenance and operation of manufacturing

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process. In addition, we need to classify the faults from a large and complex pattern processing. In this respect, we propose a recent method, one-class support vector machines (SVMs) for machine fault detection and classification.

SVMs builds on Vapnik's revolutionary findings in statistical learning theory (Vapnik, 1995). Schölkopf points out that a particular advantage of SVMs over other learning algorithms is that it can be analyzed theoretically using concepts from computational learning theory and at the same time can achieve good performance when applied to real problems (Schölkopf & Smola, 2002). The basic idea of SVMs is mapping the input data points to a high-dimensional feature space and finds a hyperplane. The algorithm is chosen in such a way to maximize the distance from the closest patterns, which is called the margin. SVMs aims to minimize an upper bound of generalization error through maximizing the margin between the separating hyperplane and data. On the contrary, the traditional methods minimize the empirical training error by mapping the input data space to high-dimensional feature data set and apply the structure risk minimization (Müller, Mika, Ratsch, Tsuda, & Schölkopf, 2001). SVMs has a good performance of classification in large data set and complex patterns processing such as text categorization (Dumais, Platt, Heckerman, & Sahami, 1998), face detection (Osuna, Freund, & Girosi, 1997), and object detection in machine vision (Roobaert & Hulle, 1999).

There are several methods to solve the one-class classification problem. Among these methods, the density estimation, the boundary methods and the reconstruction methods are noticeable. In this study, we adopt one of the boundary methods, support vector data description. Müller et al. (2001) suggested a method of adapting the SVMs methodology to the one-class classification problem (Smola, 2001). Essentially, after transforming the feature via a kernel, they treat the origin as the only member of the second class. Then using relaxation parameters they separate the image of the one class from the origin.

The purpose of this paper is to suggest a progressive method to detect and classify the machine fault signal based on one-class SVMs and to introduce a novel technique to solve this kind of problem. The proposed one-class SVMs method consists of three phases, firstly mapping input data to high-dimensional feature space. The second phase selects a kernel and computes the margin. In the final phase, the one-class classification is performed to detect the outliers. Computational experiments based on the real and artificial data set show that this approach is superior to the previous methods, of multilayer perception (MLP), which is one of the artificial neural network techniques.

The rest of this paper is organized as follows. Sections 2 and 3 present the working principles of SVMs and one-class SVMs method, respectively. Section 4 explains the application of one-class SVMs to a real machine fault detection and classification problem. Section 5 describes the computational experiments and the results of the performance evaluation. Section 6 contains concluding remarks.

2. Support vector machines

The goal of SVMs is to find optimal hyperplane by minimizing an upper bound of the generalization error through maximizing the distance, margin, between the separating hyperplane and the data. SVMs uses the preprocessing strategy in learning by mapping input space, X to a high-dimensional feature space, F . By this mapping, more flexible classifications are obtained. A separating hyperplane is found which maximizes the margin between itself and the nearest training points.

The aim of feature mapping is to find a way of computing the inner product $\langle \phi(x_i) \cdot \phi(x_j) \rangle$ in feature space directly as a function of the original input points (Cristianini & Taylor, 2000; Smola, 2001).

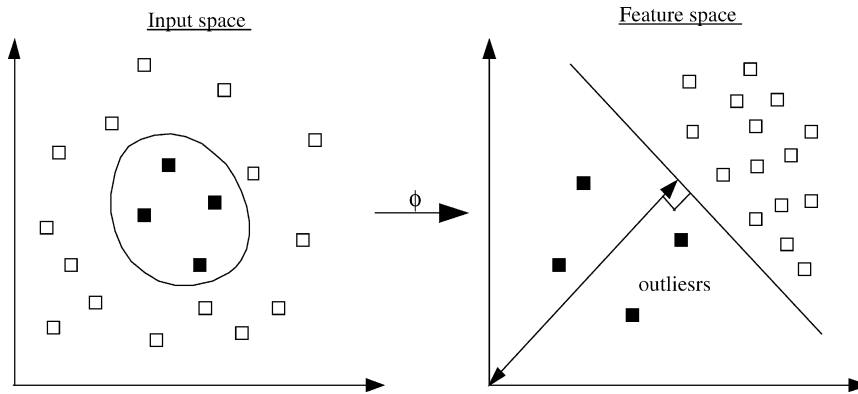


Fig. 1. Non-linear transformation from input space to high-dimensional feature space.

The feature space is very high-dimensional space where linear separation becomes much easier than input space. This is equivalent to applying a fixed non-linear mapping of the data to a feature space, in which a linear function can be used and is illustrated in Fig. 1.

Consider a pattern classifier, which uses a hyperplane to separate two classes of patterns based on given examples $S = \{x_i, y_i\}_{i=1}^m$, $y_i \in \{-1, +1\}$. The hyperplane is defined by (ω, b) , where ω is a weight vector and b a bias. Fig. 2 illustrates the concept of feature mapping and two-class separation. The SVMs solution is obtained through maximizing the margin between the separating hyperplane and the data. The linear function of hyperplane can be written as

$$f(x) = \langle \omega \cdot x \rangle + b = \sum_{i=1}^m \omega_i x_i + b \quad (1)$$

We split the problem into two steps. The first step is to fix the non-linear mapping that transforms the data into a feature space F taking advantage of the kernel theory and preserving the geometrical structure of the data into F , $x = (x_1, \dots, x_m) \mapsto \phi(x) = (\phi(x_1), \dots, \phi(x_m))$. This step is equivalent to mapping the input space X into a new space, $F = \{\phi(x) | x \in X\}$. The second step is to compute the inner product of feature space values, $\langle \phi(x_i) \cdot \phi(x_j) \rangle$. However, the computing in feature space is difficult due to the curse of dimensionality. Thus, we need to find a way of computing the inner product $\langle \phi(x_i) \cdot \phi(x_j) \rangle$ in feature space directly as a function of the original input points. We call such a direct computation method a kernel function, $K(x, y) = \phi(x) \cdot \phi(y)$. The kernel method makes it possible to merge the two steps needed to build a non-linear learning machine.

The distance between two hyperplanes is the margin of the hyperplane w with respect to the sample S . The purpose of SVMs is to maximize this distance. Using a Lagrangian, this optimization problem can be converted into a dual form that is a quadratic programming (QP) problem where the objective function is solely dependent on a set of Lagrange multipliers α_i . The QP problem is as follows

$$\min \frac{1}{2} \|w\|^2 \quad (2)$$

$$\text{subject to } y_i [\omega_i^T x_i + b] \geq 1 \quad \text{for } \forall i \quad (3)$$

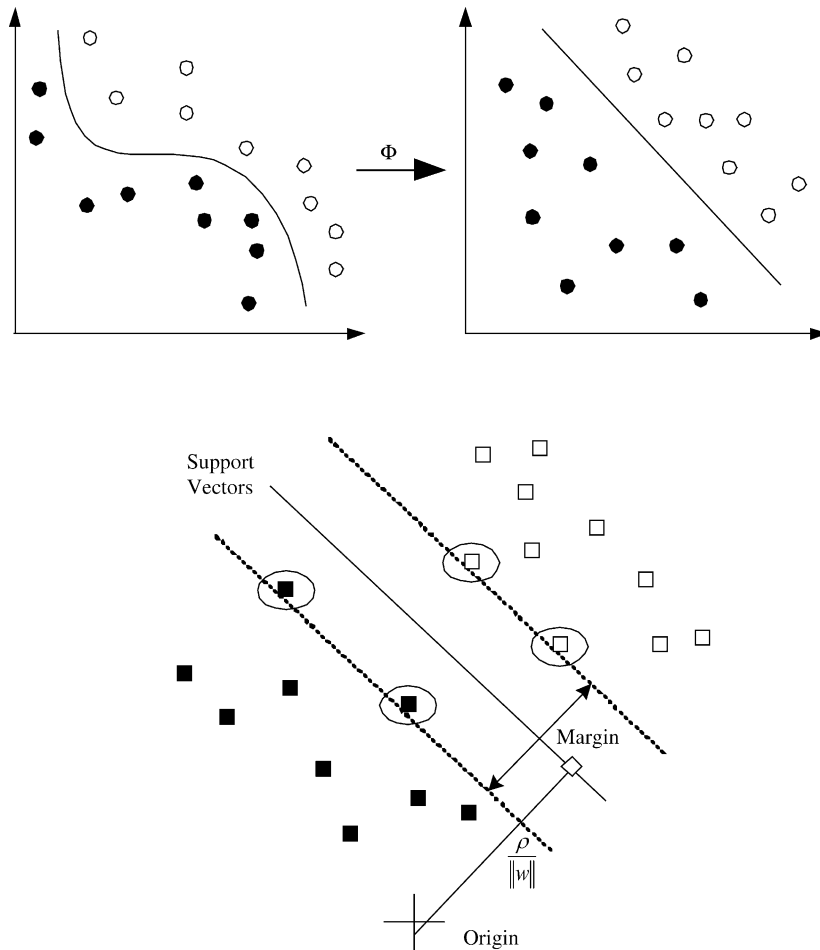


Fig. 2. Feature mapping and two-class separation.

We can get the maximal margin hyperplane with geometric margin. And then the Lagrangian is as follows:

$$L(\omega, b, \alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m \alpha_i \alpha_j y_i y_j \langle x_i \cdot x_j \rangle \quad (4)$$

subject to

$$\alpha_i \geq 0 \quad (5)$$

$$\sum_{i=1}^m \alpha_i y_i = 0 \quad \text{for all } i \quad (6)$$

The Lagrangian is has to be maximized with respect to α_i . The α_i is Lagrange multiplier and the support vectors lie only close to the hyperplane that have $\alpha_i > 0$. All others points have $\alpha_i = 0$. These support vectors contribute the computing of objective function.

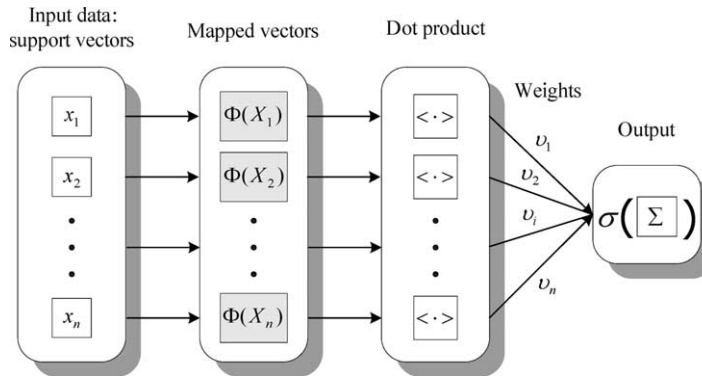


Fig. 3. Conceptual procedure for classification of SVMs.

The conceptual procedure for classification of SVMs is presented in the Fig. 3 and can be summarized as following. Applying Φ , the input data (support vectors) is non-linearly mapped into a feature space F , where dot products are executed. By the use of the kernel method, these two layers are in practice computed in one single step. The results are linearly combined by weights v_i , which can be found by solving a QP problem. Finally, the linear combination is fed into the function of σ to obtain the output.

3. One-class problem

The objective of this novel detection method is to find unusual events, clean database and distinguish typical examples from the observations of input data. Müller et al. (2001) proposed the extended SVMs methodology, one-class classification that uses only positive information. One-class classification tries to classify one class of objects, and distinguish it from all other possible objects (see Fig. 4). The objects can be classified well by the classifier, but the others will be classified as outlier. An one-class classifier should then be trained to reject this object and to label it as an outlier.

The feature space points $\phi(x_1), \dots, \phi(x_m)$ are all separable. The distance of the hyperplane is $\langle \omega \cdot \phi(x_i) \rangle \geq \rho$ and $\rho > 0$. The solution of $\min_{\omega \in F} \frac{1}{2} \|\omega\|^2$ with $\langle \omega \cdot \phi(x_i) \rangle \geq \rho$ for all i gives the unique hyperplane such that it is closer to the origin than all data and its distance to the origin is maximal among all such hyperplanes. However, not all data sets are linearly separable and it is too difficult to find a canonical hyperplanes quickly. There may be no hyperplane that splits the positive examples from negative examples.

Therefore, we give some error limits ν before the preprocessing of input data. Although this is not canonical hyperplane, it gives acceptable solutions very quickly. In the formulation above, the non-separable case would correspond to an infinite solution. By solving for a given constraint $\nu \in (0, 1]$ with slack variable ξ_i , most of the data should be separated from the origin by a large margin.

$$\min_{\omega, \xi_i, \rho} \frac{1}{2} \|\omega\|^2 + \sum_{i=1}^m (\xi_i - \nu \rho) \quad (7)$$

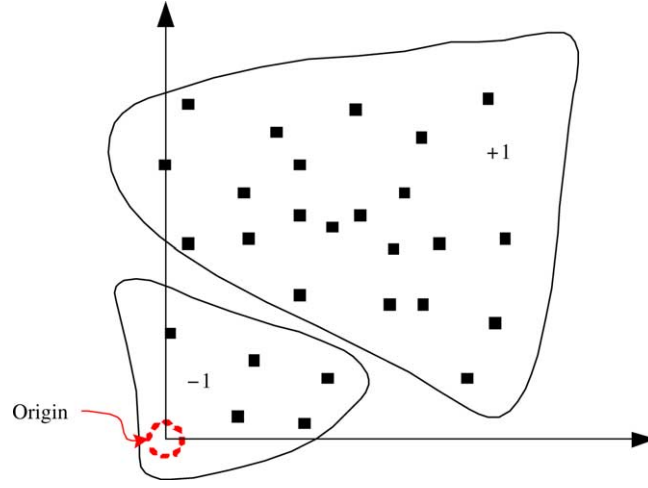


Fig. 4. Classification by one-class SVMs.

subject to

$$\langle w \cdot \phi(x_i) \rangle \geq \rho - \xi_i, \quad \xi_i \geq 0, \quad i = 1, \dots, m \quad (8)$$

Eq. (8) can be incorporated into Eq. (7) by introducing Lagrange multipliers and constructing the Lagrangian with the Lagrange multipliers $\alpha_i \geq 0$ and $\eta_i \geq 0$.

$$L(w, \xi, \rho) = \frac{1}{2} \|w\|^2 + \sum_{i=1}^m (\xi_i - \nu\rho) + \sum_{i=1}^m \alpha_i (\rho - \xi_i - \langle w \cdot \phi(x_i) \rangle) - \sum_{i=1}^m \eta_i \xi_i \quad (9)$$

For optimality, we have to compute the partial derivatives of L with respect to w , ξ , and ρ ;

$$\frac{\partial L}{\partial w} = 0, \quad w = \sum_{i=1}^m \alpha_i \phi(x_i) \quad (10)$$

$$\frac{\partial L}{\partial \xi_i} = 0, \quad \alpha_i = 1 - \eta_i \in [0, 1] \text{ then } 1 - \alpha_i - \eta_i = 0 \quad (11)$$

$$\frac{\partial L}{\partial \rho} = 0, \quad \sum_{i=1}^m \alpha_i = \nu m \quad (12)$$

Then substitute Eqs. (10) and (11) into L , using $K(x, y) = \phi(x) \cdot \phi(y)$. Finally, we obtain the dual problem of Eq. (7).

$$\min \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j K(x_i, x_j) \quad (13)$$

subject to

$$0 \leq \alpha_i \leq 1 \quad (14)$$

$$\sum_i^m \alpha_i = mv \quad (15)$$

The upper bound in inequality, Eq. (14) limits the influence of objects on the final solution. According to the KKT condition from the Lagrangian function, $\alpha_i[\rho - \xi_i - \langle w \cdot \phi(x_i) \rangle] = 0$ for all i and $\eta_i \xi_i = 0$, i.e. $(1 - \alpha_i)\xi_i = 0$. This implies as follows:

- (1) $\alpha_i \in (0, 1)$ then $\xi_i = 0$, and $\langle w \cdot \phi(x_i) \rangle = \rho$
- (2) $\xi_i > 0$ then $\alpha_i = 1$
- (3) $\langle w \cdot \phi(x_i) \rangle > \rho$ is $\alpha_i = 0$

Note that this means outliers are points x_i which fall on the wrong side of the hyperplane and support vectors are points with $\alpha_i > 0$, i.e. $\langle w \cdot \phi(x_i) \rangle \leq \rho$ outliers are included in support vectors. Consequently, ν has two properties. Firstly, at least mv points stay on or beyond the margin and at most $m(1 - \nu)$ points stay on the right side of the margin. Secondly, the fraction of outliers is equal to the fraction of support vectors. In Section 5, we examine how the ν value influences on the overall performance of one-class SVMs. As shown in the following procedural framework, the one-class SVMs can be summarized as following five steps.

3.1. One-class SVMs framework

Step 0 (*Transformation of data set*). Transform an input space into a feature space:

$$x = (x_1, \dots, x_m) \mapsto \phi(x) = (\phi_1(x), \dots, \phi_m(x)).$$

Step 1 (*Computing a margin*). Maximize the distance of the hyperplane:

$$\min_{\omega \in F} \frac{1}{2} \|w\|^2 \text{ with } \langle w \cdot \phi(x_i) \rangle \geq \rho \text{ for all } i.$$

Step 2 (*Applying $\nu \in (0, 1]$*). Calculate the trade-off between the margin and the outliers by solving the problem Eq. (7) with constraint Eq. (8).

Step 3 (*Selecting an appropriate kernel method*). Chose an appropriate kernel method: linear, radial basis function (RBF), polynomial, etc.

Step 4 (*Computing a dual problem*). Using the selected kernel method, obtain the dual problem Eq. (13) of the problem Eq. (7) and solve Eq. (13).

4. Application to machine fault detection and classification

We apply one-class SVMs discussed in Sections 2 and 3 to a real machine fault-monitoring problem. The target system for this application is an electro-mechanical device from vibration measurements. For the benchmark purpose, we used the data set collected and provided by Bergadano, Giordana,

and Saitta (1991). The brief descriptions about the target system and the data are given as following subsections.

4.1. Target system

Rotating machinery such as motor pump and ventilator is one of the most common parts in modern manufacturing industry. When a machine has rotating elements, several inevitable oscillatory motions are entailed in its components; these oscillations occur during normal machine operation. When some fault occurs in the machine, new, irregular oscillations, as well as other manifestations, come out. The fundamental aim of this predictive maintenance is to detect failures as early as possible and diagnosis their severity through an analysis of these oscillations. Therefore, being able to locate accurately the existence and severity of a fault in a machine can be of prime importance in certain areas of industry, whose machinery may be safety or emergency related.

4.2. Data description

The benchmark data set contains 209 instances. Every single instance consists of two types of information (see Table 1). One is information regarding target value (class labels) and attributes that contain machine conditions such as frequency, velocity, and phase of the harmonic components of the oscillation. The number of attributes for an instance is 8 including dummy. The other information is about machine condition classification, which is target value and can be grouped into six classes:

- (1) Problems in the joint
- (2) Faulty bearings
- (3) Mechanical loosening

Table 1

Attributes and class classification of the benchmark data set (Bergadano et al., 1991)

Attributes	0. Dummy (always 1): used for numbering 1. Class: classification (integer between 1 and 6) 2. #: component number (integer) 3. SUP: support in the machine where measure was taken (integer between 1 and 4) 4. CPM: frequency of the measure (integer) 5. MIS: measure (real) 6. MISR: earlier measure (real) 7. DIR*: filter, type of the measure and direction 8. OMEGA: RPM of the machine (integer)
Classes	1. Problems in the joint 2. Faulty bearings 3. Mechanical loosening 4. Basement distortion 5. Unbalance 6. Normal operating conditions

DIR has nine different kinds of value such as vo, ia, etc. For example, vo and ia include a set of values of <no filter, velocity, horizontal> and <filter, velocity, axial>, respectively.

Table 2

A partial example of data instance

No.	Class	Component No.	SUP	CPM	MIS	MISR	DIR	OMEGA
1	1	1	3	0	55	7.5	aa	1000
1	1	2	3	0	26	1.4	vo	1000
1	1	3	3	0	11	0.8	va	1000
1	1	4	2	0	7	5.5	ao	1000
1	1	5	2	0	25	5.5	aa	1000
1	1	6	2	0	13	0.75	vo	1000
1	1	7	2	0	9	0.95	va	1000
1	1	8	2	6000	8	0.4	io	1000
1	1	9	2	6000	5.5	0.3	ia	1000
1	1	10	2	18,000	2.75	0.6	ia	1000
1	1	11	2	20,000	6.5	0	io	1000
1	1	12	2	22,000	2.75	0	ia	1000
1	1	13	2	23,000	3.75	0	io	1000
1	1	14	2	2000	1.4	0	ia	1000
1	1	15	3	4000	18	0	io	1000
1	1	16	3	4000	1.9	0	ia	1000
1	1	17	3	3000	7	0	io	1000
1	1	18	3	24,000	1.25	0	ia	1000
1	1	19	3	20,000	4	0	ia	1000
1	1	20	3	18,000	3.5	0	io	1000
1	1	21	3	6000	2.5	0.25	io	1000
1	1	22	3	1000	12	1	io	1000
1	1	23	3	2000	4.5	0	io	1000
1	1	24	3	2000	4.5	0.35	ia	1000
1	1	25	3	4000	18	0	ma	1000

(4) Basement distortion

(5) Unbalance

(6) Normal operating conditions

These faults tend to occur together at the same time and even then, it is not always possible individuate them. Therefore, in order to make classification clear and reduce ambiguity, without loss of generality, we assume that there are no intermediate classes such as a mixed class of classes 1 and 2. Note that the above classification includes normal operation condition (class 6). For example, Table 2 shows a part of data instance. The purpose of one-class SVMs proposed in this paper is to construct a model that suggests target value of data instances in the testing set using only the given attributes.

5. Computational experiment

5.1. Comparative algorithm

The MLP, one of the artificial neural network techniques, is used for a comparative algorithm to evaluate the performance of the proposed one-class SVMs. The non-linear properties of artificial neural

networks have been regarded as suitable for applications such as machine condition diagnosis. The artificial neural networks are effective for the applications not only because the training data tend to be sparse, but also the network needs to be generalized to a certain extent. Several existing literatures have shown that a neural network has successfully recognized and classified various faults in complex condition diagnosis problems (McCormick & Nandi, 1997).

The MLP selected for a comparative algorithm in this paper consists of one hidden layer and an output layer. The network has a number of inputs determined by the number of attributes being used for classification. Each input is scaled by a constant factor that is chosen to be the largest estimate of that attribute in the training data set. The hidden layer employs a logistic activation function ($\theta(\ell) = 1/(1 + \exp(-\ell))$), where ℓ is the sum of the weighted outputs and the size of the hidden layer is determined iteratively. The hidden layer uses a linear activation function ($\theta(\ell) = \ell$). The output layer is set at six neurons according to the number of classes of machine conditions as described Section 4.2.

The target output for the network is chosen to be one for the correct class and zeros for all the other classes indicating. To achieve this, linear neuron is used. For the evaluation of the networks, the machine condition is classified as that corresponding to be the neuron with the highest value.

5.2. Configuration of one-class SVMs

One-class SVMs method has a lot of parameters to be set such as selection of kernel and the representation of the data. Moreover, it turns out to be surprisingly sensitive to specific choices of representation and kernel. In order to produce robust results, it is very important to pay attention to set these parameters. Therefore, we conducted the experiments with four different kernels using scaling method. The four kernels are as following:

- Linear: $K(x_i, x_j) = x_i^T x_j$.
- Polynomial: $K(x_i, x_j) = (\gamma x_i^T x_j + r)^d$, $\gamma > 0$.
- Radial basis function (RBF): $K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$, $\gamma > 0$.
- Sigmoid: $K(x_i, x_j) = \tanh(\gamma x_i^T x_j + r)$.

Here, γ , r , and d are kernel parameters to be set for a specific problem. The main reason to use scaling method in the training and test is to prevent attributes in greater numeric ranges from dominating those in smaller numeric ranges. Another reason is to prevent numerical difficulties during the calculation. Since most kernel values depend on the inner products of feature vectors, large attribute values might cause numerical problems. Therefore, we linearly scaled each attribute to the range [0, 1].

For the kernel parameters r and d of polynomial and sigmoid, we used the standard values. In order effectively to set the parameter $\nu \in (0, 1]$ in Eq. (7) and the kernel parameter γ , we chose values on ν and γ using cross-validation. Basically pairs of (ν, γ) are tried and the one with best cross-validation accuracy is picked. The cross-validation consists of two phases, which are broad area and intensive area. In the broad area phase, we first try to find better region with five tentative parameters, $\nu = 0.1, 0.3, 0.5, 0.7, 0.9$, and $\gamma = 0.001, 0.01, 0.1, 1, 10$. After identifying a better region, the intensive area phase can be conducted on that region. For example, if a pair of $(0.3, 0.01)$ was picked as a better region, the intensive area phase try to find best region with five parameters, $\nu = 0.250, 0.275, 0.300, 0.325, 0.350$, and $\gamma = 0.0050, 0.0075, 0.0100, 0.0125, 0.0150$. More detail description of the cross-validation will be illustrated in Section 5.4.

5.3. Training

Training both for MLP and one-class SVMs is carried out with randomly chosen 60 instances out of 209 data sets. In succession, validation is carried out both for MLP and one-class SVMs using randomly chosen 40 instances. Training is terminated when the performance on the validation data set gets worse than that of the training data set. The specific descriptions about training method for MLP and one-class SVMs are as following.

5.3.1. MLP training

The MLP is trained using a basic back propagation algorithm (Rumelhart, Hinton, & Williams, 1986) with adaptive learning and momentum. There are three aspects related to practical training of the MLP: the size of data, the number of hidden layers, and the stopping criteria.

As mentioned above, the size of input data set for training is 60, and 40 for validation. The best-hidden layers number can be obtained by trial-and-error approach. Therefore, the MLP is trained with varying sizes of hidden layer, from 2 to 14 neurons, and the best one is selected as a final configuration. The scope of hidden layers for this experiment is determined by an expression, $[\sqrt{N_{\text{inp}}N_{\text{oup}}} - 5, \sqrt{N_{\text{inp}}N_{\text{oup}}} + 5]$, where N_{inp} denotes the size of input layer (8 for this application) and N_{oup} the size of output layer (6 for this application), respectively.

When it comes to stopping criteria, we watch both training and validation errors. Typically, training error is going down and validation error first also goes down but then may begin to rise. This is the moment to stop training.

5.3.2. One-class SVMs training

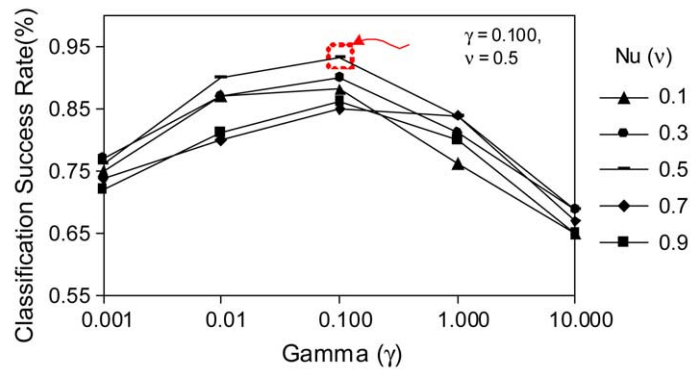
In our computational experiment, we employed the LIBSVM (version 2.5) software and modified it to train and test the one-class SVMs proposed in this paper. This is an integrated software tool for support vector classification, regression, and distribution estimation, which can handle one-class SVMs. The LIBSVM 2.5 is available at <http://www.csie.ntu.edu.tw/~cjlin/libsvm>. We conducted the following procedures to train our one-class SVMs.

- Conduct scaling on the data: linearly scaling input data to the range [0, 1].
- Select a kernel out of four kernels and find appropriate parameter values.

5.4. Results

Appropriate parameters for one-class SVMs with different four kernels are set by the two-phased cross-validation process. For example, Fig. 5 illustrates the two phases—broad and intensive area phase, sequentially for one-class SVMs with RBF kernel. As shown in the figure, final parameters for training and test were determined as $(\nu, \gamma) = (0.525, 0.075)$ and we can see that the overall performance is affected by γ more than ν . We can also recognize from this validation process that it is important to choose the appropriate parameters. In the same manner, the parameters for one-class SVMs with polynomial and sigmoid kernel were determined by cross-validation process. However, since one-class SVMs with linear kernel does not have the parameter γ , we executed the two-phased validation process only for parameter ν . The final parameters are summarized in Table 3.

(a) First phase of cross validation: broad area search



(b) Second phase of cross validation: intensive area search

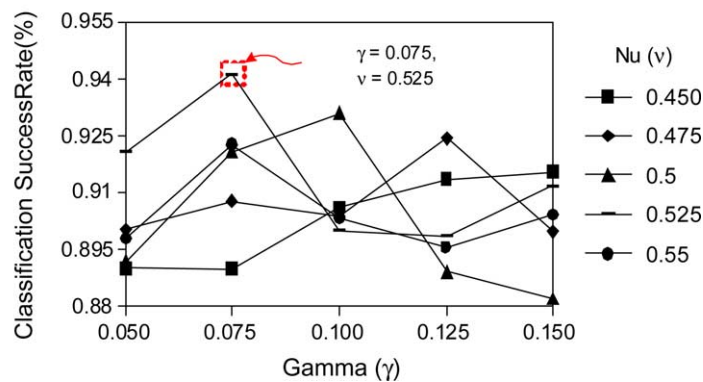


Fig. 5. Parameter selection by the two-phased cross-validation.

We evaluated the performance for the proposed one-class SVMs with different types of kernel and compared it with that of MLP in Table 4. The table shows both training and test success rates that were achieved under each different method. As can be seen, one-class SVMs with RBF kernel has consistently given the best performance of all five methods. Examining the results, we can see that MLP manages to achieve 99.7% success on the training set and 97.8% with the test set. Compared to the performance of one-class SVMs with the other kernels (linear, polynomial, and sigmoid), the MLP gave relatively good results. This means that the neural network method is robust in both the performance and less sensitivity

Table 3
Final parameter set for one-class SVMs with different four kernels

Kernel for one-class SVMs	ν	γ	r	d
Linear	0.500	—	—	—
Polynomial	0.475	0.100	0	3
RBF	0.525	0.075	—	—
Sigmoid	0.525	0.100	0	—

Table 4

Performance comparison between one-class SVMs with different four kernels and MLP

Classifier	Type of kernel	Number of classes	Number of inputs	Training success rate (%)	Test success rate (%)
MLP	–	6	209	99.7	97.8
One-class SVMs	Linear	6	209	98.7	96.0
	Polynomial	6	209	99.4	97.6
	RBF	6	209	100	99.0
	Sigmoid	6	209	99.8	98.1

to parameters. As can be seen from Fig. 5 and Table 4, however, provided the selection of the appropriate parameters and kernel is guaranteed, one-class SVMs gives better results.

In addition, we examined the performance of one-class SVMs on the various sizes of artificial data set. The artificial data were created by uniform distribution and the upper and lower bounds were selected according to the real data set. The artificial data set has five different sizes of data instances, which are 100, 200, 400, 600, and 1200, respectively. As shown in Table 5, the results do not show clear relation with the increasing size of the data set, but we can see that the performance of one-class SVMs as well as MLP is decreasing beyond 400 data. However, the performance is not linearly decreasing with the increasing size of the data set. This can be explained by the fact that the classification accuracy depends on the characteristics of data set.

6. Conclusion

We have presented one-class SVMs that finds maximum margin hyperplanes in a high-dimensional feature space, emulating Vapnik's SVMs. The objective of this paper was to show the comparableness of one-class SVMs method to artificial neural networks in the outlier detection problem of high dimensions. Experiments performed on artificial and real dataset show that the performance of this method is mostly superior to that of MLP, one of the artificial neural network methods.

We can use a variety of methods to detect and diagnosis of faults efficiently in real life. The accuracy and fastness is the barometer of reducing the cost of maintenance and operating of process plants. The proposed method, one-class SVMs served to exemplify that kernel-based learning algorithms are indeed highly competitive on a variety problems with different characteristics and can be employed as an efficient method for machine fault detection and classification.

Table 5

Performance (test success rate (%)) on the sizes of the artificial data set

Classifier	Type of kernel	Sizes of the artificial data set				
		100	200	400	600	1200
MLP	–	100	99.7	98.1	90.6	91.8
One-class SVMs	Linear	99.5	97.3	95.1	86.1	90.0
	Polynomial	100	99.9	96.8	88.6	91.6
	RBF	100	100	98.3	90.6	93.9
	Sigmoid	100	99.6	97.0	87.2	89.4

Since one-class SVMs was shown to be sensitive to the parameters and the choice of kernel, however, more reinforced consideration should be followed in how to select appropriate parameters and kernel for one-class SVMs.

References

- Bergadano, F., Giordana, A., & Saitta, L. (1991). *Machine learning: An integrated framework and its applications*. Chichester: Ellis Horwood Ltd.
- Cristianini, N., & Taylor, J. S. (2000). *An introduction to support vector machines and other kernel-based learning methods*. Cambridge, MA: Cambridge University Press.
- Dumais, S., Platt, J., Heckerman, D., & Sahami, M. (1998). Inductive learning algorithms and representations for text categorization. In *Proceedings of ACM-CIKM98, Washington, DC* (pp. 148–155).
- McCormick, A. C., & Nandi, A. K. (1997). Real time classification of rotating shaft loading conditions using artificial neural networks. *IEEE Transactions on Neural Networks*, 8(3), 748–757.
- Müller, K. R., Mika, S., Ratsch, G., Tsuda, K., & Schölkopf, B. (2001). An introduction to kernel-based learning algorithms. *IEEE Transactions on Neural Networks*, 12(2), 181–201.
- Osuna, E., Freund, R., & Girosi, F. (1997). Training support vector machines: An application to face detection. In *1997 Conference on computer vision and pattern recognition* (pp. 130–136). Puerto Rico: IEEE.
- Roobaert, D., & Hulle, V. M. (1999). View-based 3d-object recognition with support vector machines. In *1999 IEEE workshop on neural networks for signal processing* (pp. 77–84). Madison, WI: IEEE.
- Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning internal representations by error propagation. In *Parallel distribution processing* (Vol. 1). Cambridge, MA: MIT Press.
- Schölkopf, B., & Smola, A. (2002). *Learning with kernels—support vector machines, regularization, optimization and beyond*. Cambridge, MA: MIT Press.
- Smola, A. (2001). *Lecture note: Introduction to machine learning*. <http://axiom.anu.edu.au/~smola/engn4520>.
- Vapnik, V. (1995). *The nature of statistical learning theory*. Berlin: Springer.